



Problem Solving Using Programming (Java) • 26SC1101E

Restaurant Billing System

A Java Console Application

●●● team_lead.java

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Why This Project?

issues.log

Current Issues

- ✗ Manual calculation errors during peak hours.
- ✗ Slow service causing customer queues.
- ✗ Messy and illegible handwritten slips.

solution.java

Our Solution

- ✓ Automatic totals using math operators.
- ✓ Fast, repeatable digital ordering.
- ✓ Clean, itemised digital receipts.

USER FLOW

STEP 01

1. Choose Category



STEP 02

2. Add by Code & Qty



STEP 03

3. Get Final Bill

Abstract

abstract.txt

Restaurant bills are often written and added up by hand, which is slow and easy to get wrong. Our Java console program shows a menu by category, takes orders by item code and quantity, and keeps a running subtotal. The final bill lists every ordered item and adds 5% service tax and 18% GST. It also handles a wrong code, notices when nothing was ordered, and ends with a coupon message. We built it using only CO1, CO2 and CO3 concepts.

count

10

menu items

count

4

categories

count

4

methods

count

4

arrays

System Comparison

Manual Methods	Our Java System
Written on paper pads.	Fast item code entry.
Calculators and human math.	Automatic subtotals and taxes.
Errors noticed late.	Instant invalid code handling.
Messy, unclear slips.	Clean, itemised digital receipts.

Takeaway: Removes hand calculation and makes billing perfectly consistent.

Menu Data & Storage

1D Parallel Arrays

Menu stored across 4 parallel arrays: code, name, price, quantity. The same index refers to the exact same item.

Business Rules

-  4 Categories: Non-Veg, Veg, Eggitarian, All
-  Fast linear search by item code
-  5% Service Tax & 18% GST applied

// Sample of 10 Total Menu Items

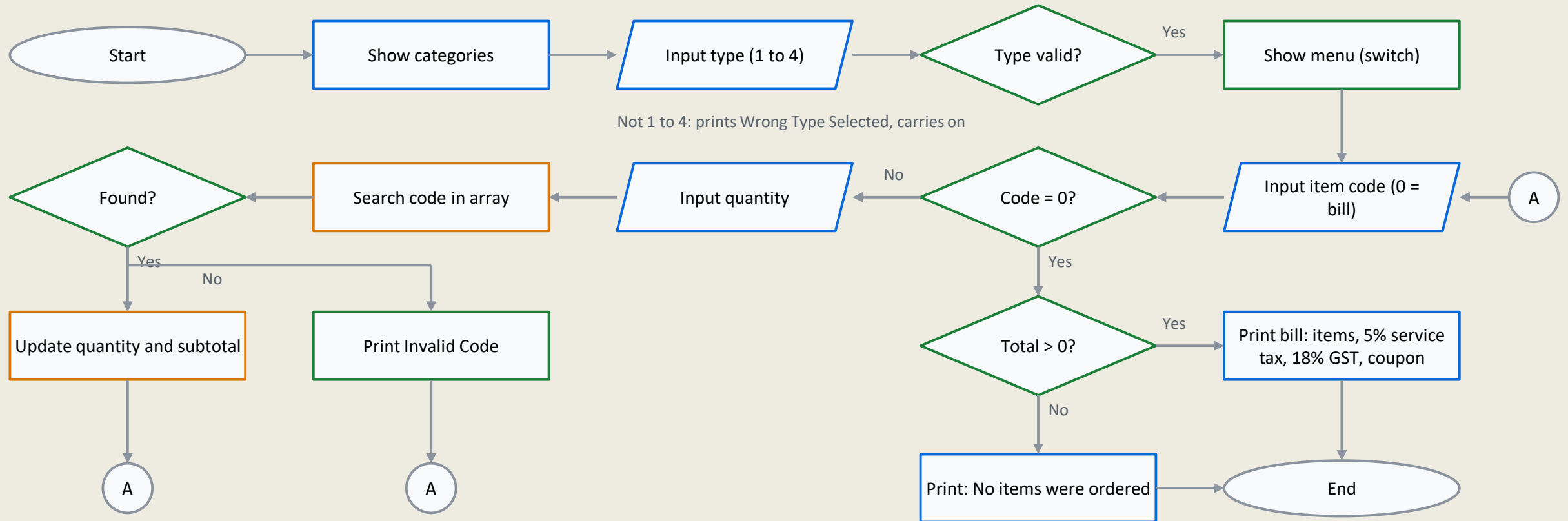
Code	Item Name	Price (Rs)
101	Chicken Biryani	220.0
201	Paneer Tikka	120.0
301	Egg Bhurji	110.0
...	+ 7 more items	...

C01: Data Types

C03: 1D Arrays

Program Flow and Topics Used

■ C01 ■ C02 ■ C03



● ● ● C01 | Basics

Algorithm and flowchart first, IPO model (input: category, code, quantity; output: bill), int, double and boolean, arithmetic for price x quantity and the taxes, Scanner input and println output.

● ● ● C02 | Control flow

Switch-case with default and break for the menu, if-else for found or invalid and total > 0, while loops for the menu, the ordering loop that ends on 0 and the search, plus a boolean flag.

● ● ● C03 | Structure

Four methods split the program into parts, four parallel 1D arrays hold the data, and we use traversal, linear search by item code and a running total.

Execution Output & Receipt Demo

```
Console_Order_Workflow.log

Enter The Type (1-4): 1
=====
                MENU
=====
101. Chicken Biryani = 220.0
102. Chicken Noodles = 120.0
103. Mutton Biryani = 240.0
104. Chicken Lollipop = 160.0
105. Chicken Tandori = 240.0
=====
Enter Item Code (or 0 to Generate Bill): 101
Enter the quantity: 2
2 x Chicken Biryani added. Subtotal: 440.0
Enter Item Code (or 0 to Generate Bill): 0
===== FINAL BILL =====
item            qty      price    final price
Chicken Biryani    2      220.0      440.0
```

```
Final_Bill_Summary.txt

Due:   Rs   541.2
You have earned a Copoun Worth 100 In Movie Ticket From Group6 theater
Coupon = Enjoy100G6
Visit Group 6 For Further Info
=====

Bill Calculation Breakdown
· Subtotal: Rs 440.0 (2 x Chicken Biryani @ 220.0)
· Service Tax (5%): Rs 22.0
· GST (18%): Rs 79.2
· Final Total Payable: Rs 541.2

Reward & Promotional Offer
· Reward: Rs 100 Movie Ticket Coupon
· Partner Theater: Group6 Theater
· Promo Code: Enjoy100G6
```

Thank You!

Presenter 01

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Upcoming Session

Next: Live output and code walkthrough

Questions & Discussion?